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TianJiu therapy for α -naphthyl isothiocyanate-induced intrahepatic cholestasis in rats treated with fresh *Ranunculus sceleratus* L.

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ABSTRACT

Ethnopharmacological relevance: TianJiu (TJ) therapy, one type of cold moxibustion, applies to specific acupuncture points with herbal patches of hot nature, providing a constant irritant to the skin until the presence of hyperemia and blistering. Traditional and clinical reports suggest that TJ is an effective therapy for the treatment of jaundice with fresh *Ranunculus sceleratus* L. (RS), in which protoanemonin is one of the main irritant constituents. However, the therapeutic effect of TJ treatment with fresh RS against intrahepatic cholestasis has not been studied in animal experiments.

Aim of the study: Present study was undertaken to investigate the effect of TJ treatment with fresh RS against intrahepatic cholestasis in rats and provide an experimental basis for the underlying mechanism of TJ therapy.

Materials and methods: Male intrahepatic cholestatic Sprague-Dawley rats induced by 2% α -naphthylisothiocyanate (ANIT, 80 mg/kg B.W.) were treated by TJ therapy with fresh RS. The levels of serum aspartate aminotransferase (AST), alanine aminotransferase (ALT), alkaline phosphatase (ALP), direct bilirubin (DBIL), total bilirubin (TBIL), total bile acid (TBA), hepatic malondialdehyde (MDA) and nitric monoxide (NO), as well as hepatic body ratio, bile flow and hepatic histopathological assay were measured and evaluated to investigate the therapeutic effect of TJ treatment with fresh RS. Phytochemical analysis of fresh and dried RS was performed by gas chromatography-mass spectrometer (GC-MS).

Results: After TJ treatment with fresh RS, the abnormally elevated levels of serum AST, ALT, ALP, DBIL, TBIL and TBA, as well as hepatic MDA and NO at 108 h were reduced significantly (versus model group, $P < 0.01$). The hepatic body ratio, bile flow and hepatic pathological change of cholestatic rats at 108 h in TJ group were restored when compared with those of model group. Thirty-one compounds including lactones, flavonoids and phenolic acids were identified and determined by GC-MS analysis. The content of protoanemonin in fresh RS (9.49%) was about 25-fold higher than that in dried RS (0.38%).

Conclusions: TJ treatment with fresh RS exhibited good therapeutic effect on ANIT-induced intrahepatic cholestasis in rats, which may be due to the attenuated oxidative stress in the liver tissue. It is rational for the ancients to choose fresh RS as TJ herbal patches because of its abundant protoanemonin with the character of irritant. The qualitative and quantitative results of GC-MS analysis provided the chemical basis of TJ therapy with fresh RS, which can be regarded as a simple and efficient method for the treatment of cholestasis hepatitis.

1. Introduction

Jaundice is a kind of symptom caused by bilirubin metabolic disorder in the body. It is clinically characterized by the yellow sclera, yellow mucous membrane, yellow skin and other tissues being stained yellow. Many pathogenic factors lead to the occurrence of jaundice, among which intrahepatic cholestasis is proven to be the most important one. Hepatocyte dysfunction or bile duct obstruction blocked

normal bile flow and resulted in cholestasis (Arrese and Trauner, 2003). Intrahepatic cholestasis originally caused liver damage followed by biliary fibrosis, cirrhosis and finally end-stage liver disease (Guicciardi and Gores, 2002; Boyer, 2007). Only a few drugs exhibited the therapeutic applications in intrahepatic cholestasis, some of which had strong drug dependences and other side effects (Neri et al., 2002; Bacq et al., 2012). Up to now, ursodeoxycholic acid (UDCA) is the only drug for cholestasis approved by FDA, and has been widely used for the

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treatment of a variety of cholestasis related liver diseases including primary biliary cirrhosis, primary sclerosing cholangitis, intrahepatic cholestasis of pregnancy, drug induced liver injury, and cystic fibrosis (Fickert et al., 2001; Roma et al., 2011; Jansen et al., 2017). However, some patients have limited or even no response to UDCA therapy. In addition, high doses of UDCA led to side effects such as osteoporosis (Ghonem et al., 2015). Modern pharmacological studies confirmed that traditional Chinese medicine (TCM) had a good therapeutic effect on cholestasis (Ma et al., 2018) and possessed several positive features including higher efficiency, less adverse reactions and lower cost than chemical drugs (Dhiman and Chawla, 2005). Therefore, it is of vital importance to find a new TCM for the treatment of cholestasis.

Tianjiu (TJ) therapy as an external application is also known as “cold moxibustion” or “vesiculating moxibustion”. It was first recorded in ancient Chinese medical works *Wushi'er Bingfang* that was discovered in Mawangdui in a tomb in 1973 and sealed in 168 BCE under the Han dynasty. TJ therapy was applied to the specific acupuncture points with selected irritant fresh herbal patches, which provided a constant irritant to the skin until the presence of hyperemia and blistering, thus irritating the related meridians and harmonizing the targeted visceral system (Chan et al., 2015). In addition, TJ treatment could invigorate *qi* movement, harmonize nutrient absorption and defense mechanisms, thereby resolving stagnation in the body and stasis of the blood (Kun et al., 2016). TCM has a long history of using TJ as a unique therapy to treat allergic diseases (Lee et al., 2009; Schafer et al., 2002). Besides its irritant effect on acupuncture points or affected areas, the direct pharmacological effect of TJ treatment was similar to that of transdermal administration. Clinical and basic researchers found that TJ treatment, a regulator of anti-inflammatory cytokines, could regulate the neuro-immunological system (Brinkhaus et al., 2004; Li et al., 2007; Zou et al., 2007). Both irritant and pharmacological effects of fresh herbal patches on the acupuncture points contributed to the unique advantage of TJ therapy (Huang et al., 2018). Accordingly, TJ treatment is an excellent combination of drug and acupuncture without needle.

Ranunculus sceleratus L. (RS), one of herbaceous medicinal plants of Ranunculaceae family, is especially common in slow ditches and muddy-bottomed ponds of mineral-rich water and distributed throughout the world. Compendium of Materia Medica written by Shizhen Li in Ming Dynasty of ancient China indicated that fresh RS was rubbed and then pasted onto the acupuncture point of Cunkou overnight, thus blistering at the skin surrounding Cunkou and feeling of fire scald, which was named as “vesiculating moxibustion” and used to cure the jaundice induced by malaria (Li, 1977). It was reported that genus *Ranunculus* plants exhibited the pharmacological activities of liver protection and anti-inflammatory, which were attributed to the presence of flavonoids, lactones and phenolic acids (Saber et al., 1968; Cao et al., 1992; Li et al., 2005; Akkol et al., 2012; Kong et al., 2018). Protoanemonin, as one of the lactones and a strong blistering agent, was primarily responsible for the characteristic topical irritant of fresh RS and translated into nonirritating anemonin by dimerization. All parts of the plant were irritant when fresh, but the irritant character of fresh RS was destroyed when the plant was heated or dried (Prieto et al., 2003). Therefore, fresh RS is the first choice as the TJ herbal patches in the traditional and modern clinical application, whereas the dried RS is an alternative. Up to now, there are 426 cases of clinical reports on the treatment of hepatobiliary diseases, such as chronic hepatitis, infectious hepatitis, jaundice and cholestasis, by TJ therapy with fresh genus *Ranunculus* plants (Zhu, 1995; Li and Zhen, 1987; Yan, 1986). However, the therapeutic effect of TJ treatment with fresh RS has not been studied in animal experiments.

In this study, fresh RS patches were pasted onto specific acupuncture points of ANIT-induced liver injury rats to observe the therapeutic effect of TJ treatment. The levels of serum enzymes and components as well as hepatic MDA and NO were detected to evaluate the liver function of rats. The body weights, liver weights and hepatic body

ratios were recorded to examine the living condition of rats. Histological examination and bile secretion were performed to assess the morphological change and recovery degree of liver injury in cholestatic rats. Finally, GC-MS analysis was studied to explore the potential active components of fresh and dried RS. To the best of our knowledge, it is the first time that the therapeutic effect of TJ treatment with fresh RS on cholestatic rats was investigated.

2. Materials and methods

2.1. Chemicals

α -Naphthylisothiocyanate (ANIT) was purchased from Sigma-Aldrich, Inc. (St. Louis, MO). Ursodeoxycholic acid (UDCA) was acquired from Losan pharma GmbH Co., Ltd. (Germany). The assay kits for the determination of aspartate transaminase (AST), alanine aminotransferase (ALT), alkaline phosphatase (ALP), total bilirubin (TBIL), direct bilirubin (DBIL), total bile acid (TBA) and malondialdehyde (MDA) were purchased from Jiancheng Institute of Biotechnology (Nanjing, China). The assay kit for the determination of nitric monoxide (NO) was purchased from Beyotime Biotechnology (Shanghai, China). Ultrapure water was produced using Milli-Q system (Millipore, Bedford, MA, USA). Other chemicals were of analytical grade and commercially available.

2.2. Animals

Male Sprague Dawley rats (200 $\pm$ 20 g) were supplied by the Laboratory Animal Center of Anhui Medical University (Hefei, China). Rats were housed in stainless-steel cages in a room with controlled temperature (25 $\pm$ 2 °C) and relative humidity (55 $\pm$ 5%) and freely available food and water. Rats were allowed to acclimatize to the housing conditions for one week and then fasted with free access to water 12 h prior to the experiments. The study and all protocols were approved by the Committee of Ethics of Animal Experimentation of Anhui Medical University.

2.3. Plant material

The batch of *Ranunculus sceleratus* L. (No.: 20180428) was collected in Hefei Shushan National Forest Park (Hefei, China) and preserved in the dark at -20°C . It was authenticated by Professor Huasheng Peng (School of Pharmacy, Anhui University of Chinese Medicine). A voucher specimen of the plant is kept for reference in the herbarium of Anhui Medical University (Hefei, China).

2.4. GC-MS analysis of *Ranunculus sceleratus* L

5 g fresh or dried RS was ground into powder and extracted by ultrasonic extraction with diethyl ether (1:2, w/v) for 30 min. Extraction fluid was filtered through a 0.22 μm Millipore membrane. The filtrated solution was injected directly for GC-MS analysis.

The chemical constituents of RS were analyzed by GC-MS using a QP2010SE chromatographic system (Shimadzu Corporation, Kyoto, Japan). The temperature program for the VB5 column (30 m $\times$ 0.25 mm inner diameter, 0.25 μm film thickness) was set at an initial temperature of 50 °C for 1 min, a rate of temperature increase of 8 °C/min and final temperature of 160 °C for 2 min, then temperature increase of 8 °C/min and final temperature of 280 °C for 15 min. The carrier gas was He (99.999%) with a constant flow at 1.0 mL/min. 1 μL of sample solution was injected in the 1:10 split mode and analyzed using both the full scan (m/z 40–600) and the electron impact mode in the MS at the ionization energy of 70 eV. The temperatures at the inlet and MS transfer line were set at 230 °C and 280 °C, respectively. The analytes were identified using the NIST 17 Mass Spectral Database and Wiley data library.

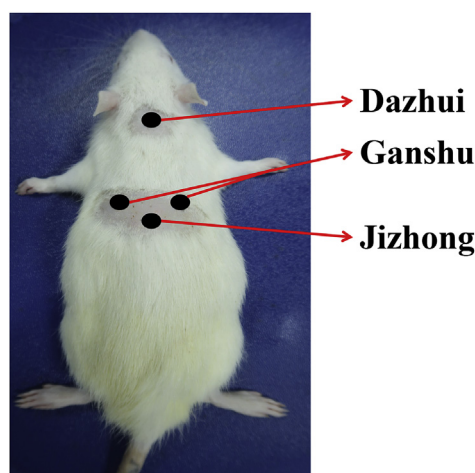


Fig. 1. Location of rat's acupuncture points of Dazhui, Ganshu and Jizhong.

2.5. Experimental design

Rats were divided into four groups randomly with eight in each group as follows: (1) Normal group: rats given with plant oil; (2) Model group: rats given with ANIT (80 mg/kg B.W.); (3) TJ group: rats given with ANIT (80 mg/kg B.W.) and treated with fresh RS patches; (4) UDCA group: rats given with ANIT (80 mg/kg B.W.) and treated with UDCA (100 mg/kg B.W.). ANIT was dissolved in plant oil to a final concentration of 2% beforehand.

Rats were kept in an immobilization apparatus without anesthesia, a system both convenient for TJ treatment and comfortable to experimental rats for reducing stress (Tu et al., 2012). Fresh RS was smashed into herbal patches, which were applied on the acupuncture points of Dazhui (GV14), Ganshu (BL18) (both sides) and Jizhong (GV6) (Fig. 1) for TJ sticking therapy (Hua et al., 1991; Fang, 1993). Herein, BL18 refers to the 18th point in bladder meridian, whereas GV14 and GV6 refer to the 14th point and 6th point in Governor Vessel, respectively. The selected acupuncture points are commonly applied in the clinical acupuncture for the effective treatment of hepatobiliary diseases (Yang, 2006). After rats were anesthetized, hair around the acupuncture points (1.5 cm × 1.5 cm) was shaved, and then rats were unhaird with Na₂S (80 g/L) for 3 min. In the next five days, fresh RS patches with the size of soybean were applied to the acupuncture points and covered with medical adhesive tapes. Each TJ treatment lasted for 12 h per day. Rats in UDCA group were orally administered twice per day for five consecutive days.

Blood samples were collected from the orbital venous of rats at 0 h, 36 h, 60 h, 84 h and 108 h after modeling for the detection of serum biochemical indicators. Subsequently, all the rats were anaesthetized with urethane (1.5 g/kg B.W., intraperitoneal) to carry out the experiment of bile excretion for 2 h. After the animals were sacrificed, the livers were removed for the evaluation of histological examination and the assays of hepatic MDA and NO.

2.6. Biochemical determinations

2.6.1. Assays of the levels of serum enzymes and components

After standing for 1 h at room temperature, the blood samples of rats were centrifuged at 4000 rpm for 10 min at 4 °C to collect the supernatants. The levels of serum ALT, AST, ALP, TBIL, DBIL and TBA were used as the indices of hepatic cell injury and measured by commercial test kits from Nanjing Jiancheng Bioengineering Institute (Nanjing, China).

2.6.2. Assays of the levels of hepatic MDA and NO

Each piece of liver from rats was homogenized with homogenizer

and centrifuged, and the obtained supernatant was used immediately for the assays of hepatic MDA and NO by commercially available test kits from Nanjing Jiancheng Bioengineering Institute (Nanjing, China). Protein concentrations were determined according to the Bradford method, using known concentrations of albumin as the standard. The procedures conformed to the instructions from related kits.

2.7. Experiments of bile excretion

Rats were anaesthetized with urethane (1.5 g/kg B.W., intraperitoneal) and kept under warming shelf to maintain rats' body temperature. The bile duct was cannulated with a PTFE tube (inner diameter, 0.3 mm; outer diameter 0.5 mm; ShengChen Co, Shanghai, China), which were eligible to insert into bile duct to collect the bile for 120 min. Bile flow was calculated by the weight of each specimen.

2.8. Histological examination

After biliary drainage, liver tissue obtained from the same lobe of liver in each rat was immediately removed and fixed in 4% neutral buffered formalin. The liver tissue samples were dehydrated in alcohol, embedded in paraffin, and then sectioned at 5–7 μm. The sections were stained by hematoxylin and eosin (H&E) for histological assessment and observed under a light microscopy (Leica DM6 B, Germany).

2.9. Statistical analysis

Statistical analysis was performed using SPSS 16.0. One-way analysis of variance (ANOVA) followed by a LSD-t-test was used for multi-group comparisons. All the data were expressed as the mean ± standard deviation (SD). The differences were considered statistically significant when the value of *P* was less than 0.05.

3. Results

3.1. Chemical constituents of RS by GC-MS analysis

The GC-MS chromatograms and chemical constituents of fresh and dried RS are shown in Fig. 2 and Table 1, respectively. Thirty-one compounds were identified by the evaluation of MS spectra with those in NIST 17 and Wiley data library, in which lactones, flavonoids and phenolic acid were the main constituents of the total peak area. The detailed results of GC-MS analysis indicate that the contents of protoanemonin (peak 4), 1,3-diphenyl-1-butene (peak 13), cis-aconitic anhydride (peak 14), 3-hydroxy-5-methoxyflavone (peak 17), 2,2-methylenebis(6-tert-butyl -4-methylphenol) (peak 20) and phenacetic acid (peak 29) were 9.49%, 18.2%, 27%, 4.93%, 5.33% and 22% in fresh RS and 0.38%, 3.49%, 0.39%, 1.39%, 11.28% and 21.92% in dried RS, respectively. It is worth noting that the content of protoanemonin (with the character of irritant) in fresh RS was about 25-fold higher than that in dried RS.

3.2. Effect of TJ treatment with fresh RS on living conditions

Rats in normal group were viable with clear urine and velvet and glossy hair, the bile could be easily drained. As a comparison, ANIT-induced rats were inactive and anorectic with deep yellow urine and hair in a mess, and the amount of drained bile decreased. After treatment with TJ and UDCA, the living conditions of ANIT-induced rats were improved. As displayed in Table 2, compared with those of normal rats, the body weights of rats after exposure to ANIT decreased significantly (*P* < 0.01), and the liver weights and hepatic body ratios of model group increased significantly (*P* < 0.01). The liver weights and hepatic body ratios of UDCA group and TJ group obviously decreased (*P* < 0.01), although a significant (*P* < 0.05) and no significant increase of body weight were observed in UDCA group and TJ group

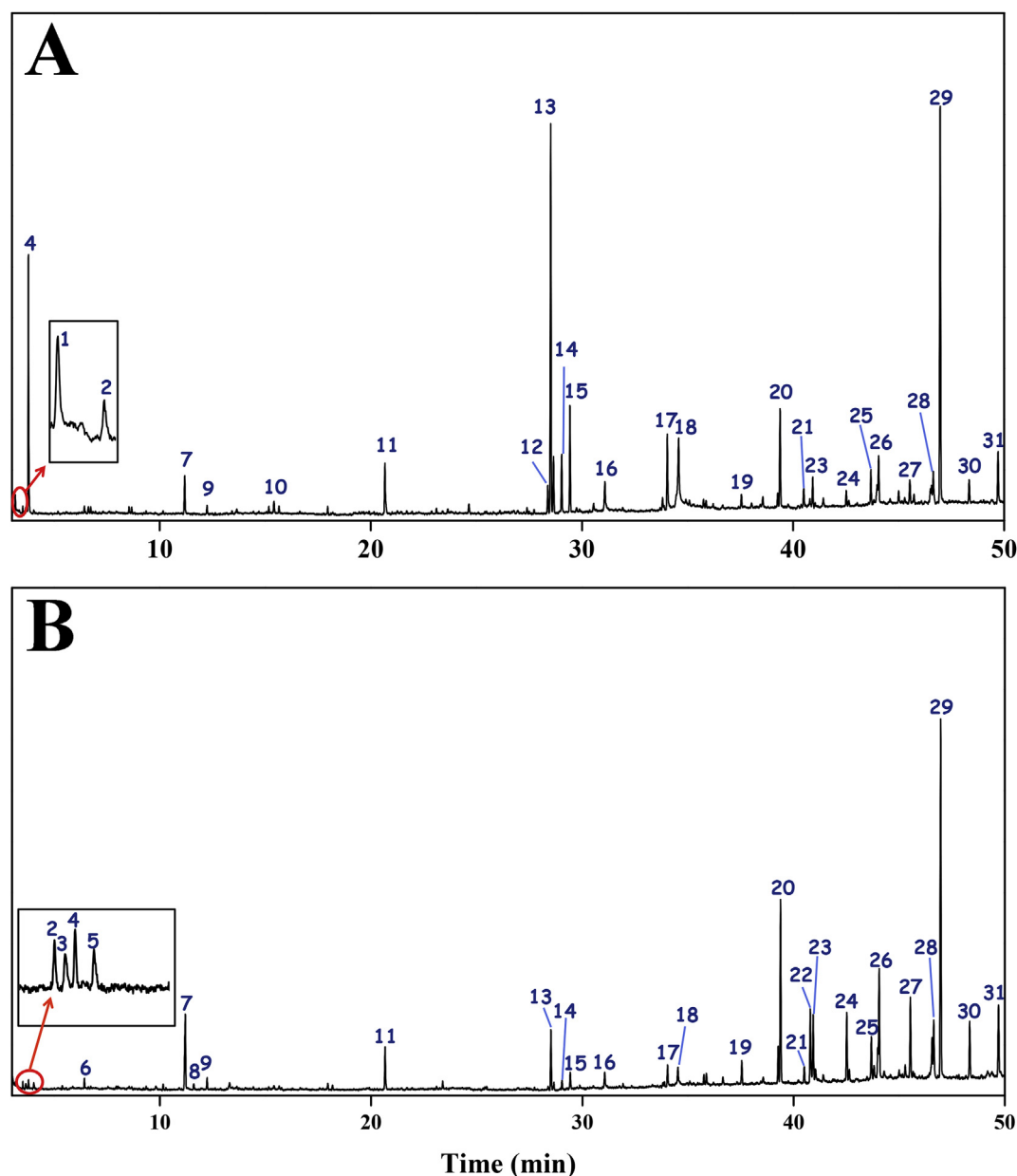


Fig. 2. GC-MS Chromatograms of fresh RS (A) and dried RS (B).

when compared with those of model group.

3.3. Effect of TJ treatment with fresh RS on the levels of serum enzymes and components

The levels of serum enzymes and components of rats at different time points (0 h, 36 h, 60 h, 84 h and 108 h) were analyzed. Fig. 3 indicates that rats in model group were poisoned by ANIT with the levels of serum enzymes increasing significantly and reaching the highest at 36 h. After treatment with TJ and UDCA, the levels of serum enzymes decreased obviously, indicating that the injury caused by ANIT was gradually recovered, and two treatment groups could achieve certain therapeutic effects.

Rats administered with ANIT exhibited liver injury and cholestasis by significant elevation of indicators in serum samples for hepatic cell damage (ALT and AST) and cholestasis (ALP, TBIL, DBIL and TBA). As shown in Fig. 4, the levels of serum ALT, AST, ALP, TBIL, DBIL and TBA at 108 h increased significantly ($P < 0.01$) in model group when compared with those of normal group, manifesting that acute

cholestasis and liver injury were induced by 2% ANIT (80 mg/kg B.W.) administration in rats. When ANIT-induced rats were treated with TJ therapy, the levels of serum ALT, AST, ALP, TBIL, DBIL and TBA decreased significantly (versus model group, $P < 0.01$). In addition, UDCA group presented a protective effect on liver injury, which was similar to TJ group.

3.4. Effect of TJ treatment with fresh RS on the levels of hepatic MDA and NO

The exposure to ANIT markedly increased the levels of MDA and NO in the liver tissues (versus normal group, $P < 0.01$), indicating that the lipid peroxidation of liver injury was induced by 2% ANIT (80 mg/kg B.W.) (Fig. 5). TJ treatment significantly decreased the elevated levels of hepatic MDA and NO when compared with model group ($P < 0.01$). Additionally, it was found that UDCA group had a significant protective effect on the lipid peroxidation of liver injury induced by ANIT ($P < 0.01$).

Table 1
Chemical constituents of fresh and dried RS detected by GC-MS.

No	RT (min)	Constituent	Fresh (%)	Dried (%)
1	3.1	1,3-Isobenzofurandione	0.52	–
2	3.5	Benzeneethanol	0.22	0.32
3	3.6	p-Xylene	–	0.39
4	3.7	Protoanemonin	9.49	0.38
5	4.1	2H-Inden-2-one	–	0.4
6	6.3	Chalcone	–	0.44
7	11.2	(S)-5-Hydroxymethyl-2(5H)-furanone	2.13	5.76
8	11.6	5-(Hydroxymethyl)-dihydro-2(3H)-furanone	–	0.33
9	12.2	2,5-Furandione	0.32	0.55
10	15.3	4-Isopropyl-1,3-cyclohexanedione	0.58	–
11	20.2	Butylated Hydroxytoluene	2.74	2.49
12	28.3	2-Propanedioic acid	1.5	–
13	28.5	1,3-Diphenyl-1-butene	18.2	3.49
14	29.1	cis-Aconitic anhydride	27	0.39
15	29.4	Methyl-3-phenoxybenzoate	4.88	1.02
16	31.3	2-Propylmalonic acid	2.19	1.02
17	34.1	3-Hydroxy-5-methoxyflavone	4.93	1.39
18	34.5	1,6-Dihydrocarveol	4.77	1.08
19	37.5	Butanoic acid	0.71	1.35
20	39.5	2,2'-Methylenebis(6-tert-butyl-4-methylphenol)	5.33	11.28
21	40.5	1-Cyclohexene-1-carboxylic acid	1.03	0.96
22	40.7	2H-1-Benzopyran-7-ol	–	4.98
23	40.9	Cyclohexanecarboxaldehyde	1.43	4.01
24	42.5	3-Phenyl-1-indanone	0.72	4.09
25	43.1	(2-Acetyl-5-methyl-cyclopentyl)-acetic acid	1.76	3.05
26	44.5	Trtriacontane	2.34	8.33
27	45.2	3-(2-Hydroxycyclopentylidene)-2-methylpropionyl-c acid	1.44	4.87
28	46.6	1-Heptacosanol	1.45	3.74
29	47.4	Phenacetic acid	22	21.92
30	48.2	Butanedioic acid	1.23	3.62
31	49.3	Hydroxyvaleric acid	2.9	5.09

Table 2
Effect of TJ treatment with fresh RS on body weights, liver weights and hepatic body ratios of ANIT-induced rats (mean $\pm$ SD, n = 8).

Group	Body weight	Liver weight	Hepatic body ratio
Control	236.6 $\pm$ 21.68	7.41 $\pm$ 0.74	3.15 $\pm$ 0.39
Model	168.86 $\pm$ 11.07 ^{##}	9.72 $\pm$ 1.04 ^{##}	5.80 $\pm$ 0.86 ^{##}
TJ	172.88 $\pm$ 12.02	7.89 $\pm$ 1.42 ^{**}	4.56 $\pm$ 0.72 ^{**}
UDCA	187.88 $\pm$ 7.49 [*]	7.63 $\pm$ 1.02 ^{**}	4.08 $\pm$ 0.66 ^{**}

^{##}P < 0.01 versus normal group.

^{*}P < 0.05.

^{**}P < 0.01 versus model group.

3.5. Effect of TJ treatment with fresh RS on the excretion of bile

Bile flow is a critical index of intrahepatic cholestasis induced by 2% ANIT (80 mg/kg B.W.). As shown in Fig. 6, the bile flow in ANIT-induced intrahepatic cholestasis rat with the bile excretion of $0.67 \pm 0.07 \mu\text{L}/\text{min}/100 \text{ g B.W.}$ in model group decreased significantly ($P < 0.01$) when compared with that of normal group ($1.91 \pm 0.09 \mu\text{L}/\text{min}/100 \text{ g B.W.}$). After treatment with TJ and UDCA, the bile flow of ANIT-induced intrahepatic cholestasis rat significantly increased to $0.99 \pm 0.05 \mu\text{L}/\text{min}/100 \text{ g B.W.}$ in TJ group ($P < 0.01$) and $1.15 \pm 0.05 \mu\text{L}/\text{min}/100 \text{ g B.W.}$ in UDCA group ($P < 0.01$), respectively.

3.6. Effect of TJ treatment with fresh RS on histological changes of liver

Histological evaluation can provide the direct evidence of therapeutic effect of TJ treatment with fresh RS on ANIT-induced cholestatic liver injury. As shown in Fig. 7, hepatic tissue in normal group

exhibited normal cell structure with obvious formation of hepatic lobule, orderly hepatocyte cord, large nuclear and round hepatocyte nucleus and evident nuclear membrane (Fig. 7A). In comparison, hepatic tissue in model group showed loss of cell boundaries, necrosis and degradation changes in severe inflammatory cell infiltration (Fig. 7B). The severity of neutrophil infiltration and other histological damage induced by 2% ANIT (80 mg/kg B.W.) in TJ group (Fig. 7C) was obviously reduced when compared with that of model group. The H&E-stained hepatic tissue of UDCA group exhibited a mild degree of hepatocyte degeneration with less inflammatory cell infiltration (Fig. 7D).

4. Discussion

Liver is responsible for the selective uptake, metabolism and excretion of drugs, xenobiotic and environmental toxins, which makes liver susceptible to drug toxicity. In epidemiological studies, cholestasis and hepatocyte injury are the clinical manifestations of drug-induced hepatotoxicity (Björnsson and Olsson, 2005; Real et al., 2019). So far, the treatment of cholestasis related liver diseases is limited, and only UDCA can reduce the absorption of cholesterol and improve the life quality of patients. Therefore, some new therapeutic strategies for the treatment of cholestasis are expected.

TJ therapy is a conventional method for treating diseases in TCM, especially for the patients who cannot be orally administered. According to the TCM theory, each herbal medicine has a specific association with the meridians. Applying the selected herbal patches of hot nature on the appropriate acupuncture points or the diseased part could stimulate the given meridians, thereby regulating the visceral functions connected with given meridians (Kun et al., 2016). For example, TJ treatment associated with the liver meridian will act upon liver diseases. With the help of the irritant constituents of herbal patches of hot nature, TJ treatment is expected to open the pores of the skin, speed the blood flow and rebalance the immune system (Chan et al., 2015). It was reported that the mechanism of TJ therapy included inhibiting the release of mast cell mediators and modulating the levels of serum IgE as well as lymphocyte and/or macrophage activity (Xue et al., 2004).

The ancient Chinese medicine records show that fresh RS used as TJ herbal patches is more effective than dried one. In this study, a comparative study on the chemical constituents of fresh and dried RS by GC-MS analysis was performed. Protoanemonin, as one of the lactones, is an acknowledged strong blistering agent in the Ranunculaceae family and primarily responsible for the characteristic topical irritant of fresh RS (Prieto et al., 2003). Interestingly, the content of protoanemonin with the character of irritant in fresh RS (9.49%) was about 25-fold higher than that in dried RS (0.38%). In addition, the main constituents of RS were identified as flavonoids, lactones and phenolic acids, which possessed the bioactivities of anti-inflammation, anti-tumor, liver-protecting and modulating cell membrane fluidity and function (Prieto et al., 2003; Wang et al., (2009); Zhong et al., 2011; Yan et al. (2015)). It was found that fresh RS exhibited a good therapeutic effect on ANIT-induced cholestasis rats. The abundant protoanemonin with the character of irritant in fresh RS caused skin inflammation and injury of mucous membranes and enhanced the permeability of cell membrane, thus contributing to the regulation of body's immune system and the improved absorption of lactones, flavonoids and phenolic acid into the body for the treatment of cholestasis jaundice (Kocak et al., 2016; Ahmed, 2011). However, dried RS with limited protoanemonin just caused skin a slight injury. It was reported that some patients with biliary stagnation had no response to UDCA (Beuers et al., 2015; Ali et al., 2016). Therefore, TJ therapy with fresh RS will be an effective strategy for the treatment of liver diseases, especially for cholestasis hepatitis.

ANIT is an indirect hepatotoxic agent and causes liver injury mainly through membrane lipid peroxidation, which further leads to the degeneration and necrosis of hepatocytes, as well as swelling and necrosis

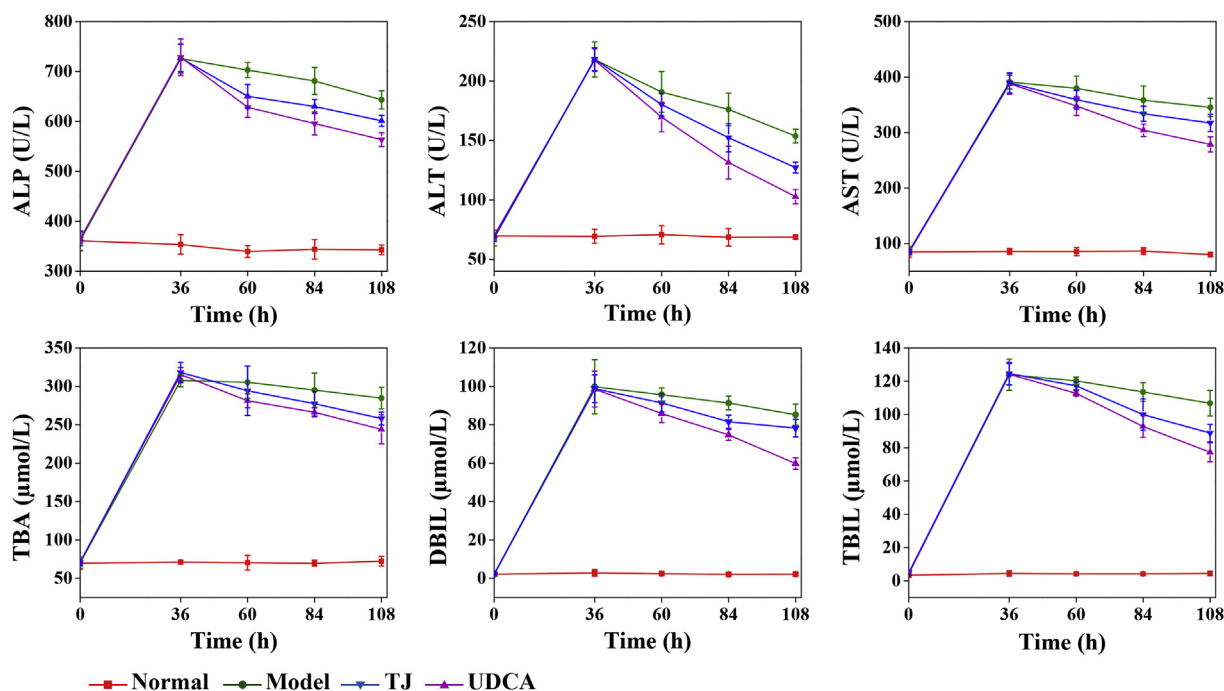


Fig. 3. Levels of serum ALP (A), ALT (B), AST (C), TBA (D), DBIL (E) and TBIL (F) in normal, model, TJ and UDCA groups at different time points (0 h, 36 h, 60 h, 84 h and 108 h). Data represent mean $\pm$ SD (n = 8 for each group).

of bile duct epithelial cells. The peak of ANIT-induced liver injury, which includes the obstruction of bile ducts, the formation of significant cholestasis, and hepatic parenchymal cell damage dominated by punctiform necrosis, was found between 24 h and 48 h in rats (Kossor et al., 1995). In our preliminary experiment, it was found that the peak time of serum levels of AST and TBIL in ANIT-induced rats after modeling occurred at 36 h, and then blood samples were collected every 24 h for the detection of serum biochemical indicators. The model of ANIT-induced intrahepatic cholestasis is simple and easy to replicate, and the biochemical indicators and pathological changes induced by liver damage are similar to those in human intrahepatic cholestasis

(Waters et al., 2001). Accordingly, ANIT-induced intrahepatic cholestasis and liver dysfunction in rats were applied to investigate the therapeutic effect of TJ therapy with fresh RS.

The levels of serum enzymes (ALT, AST and ALP) and components (TBA, TBIL and DBIL) are important indexes for the clinical manifestation of cholestasis hepatitis. ALT and AST are transaminase enzymes associated with hepatocytes. After the plasma membrane of a liver cell is damaged, the enzymes usually appear when the cytosol leaks into the bloodstream. Therefore, the levels of serum ALT and AST are sensitive markers for the diagnosis of liver disease. ALP is an enzyme found in cells in the inner wall of liver and biliary tract and used to evaluate

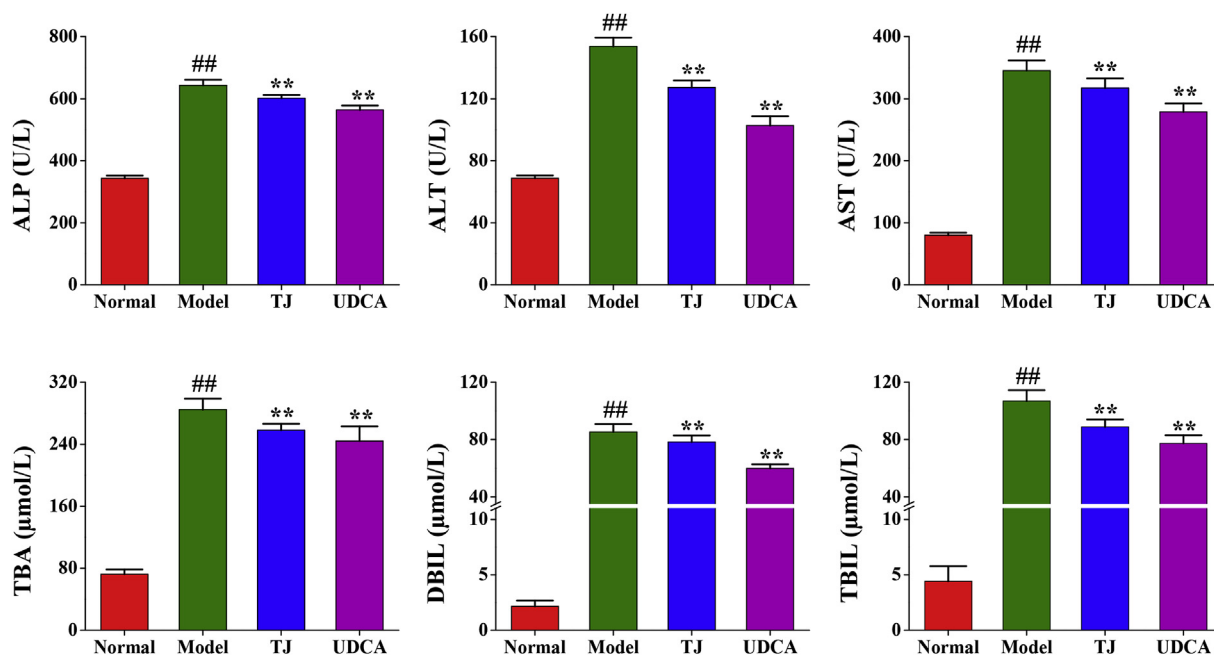


Fig. 4. Effect of TJ treatment with fresh RS on the levels of serum ALP (A), ALT (B), AST (C), TBA (D), DBIL (E) and TBIL (F) at 108 h after 2% ANIT (80 mg/kg B.W.) administration in rats. Data represent mean $\pm$ SD (n = 8 for each group). ##P < 0.01 versus normal group; **P < 0.01 versus model group.

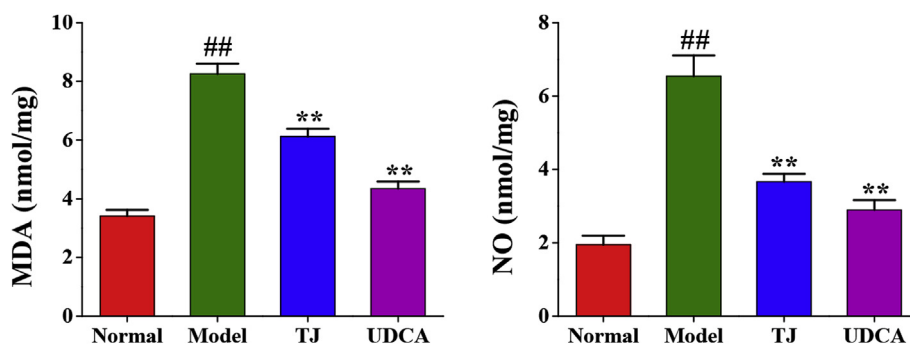


Fig. 5. Effect of TJ treatment with fresh RS on the levels of hepatic MDA (A) and NO (B) at 108 h after 2% ANIT (80 mg/kg B.W.) administration in rats. Data represent mean $\pm$ SD (n = 8 for each group). ^{##}P < 0.01 versus normal group; ^{**}P < 0.01 versus model group.

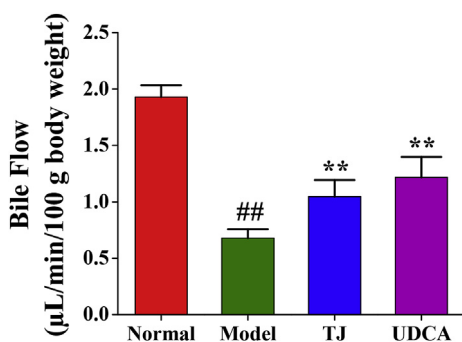


Fig. 6. Effect of TJ treatment with fresh RS on the excretion of bile at 108 h after 2% ANIT (80 mg/kg B.W.) administration in rats. Data represent mean $\pm$ SD (n = 8 for each group). ^{##}P < 0.01 versus normal group; ^{**}P < 0.01 versus model group.

biliary tract injury in the liver function tests. Generally, ALP is the first test for biliary tract disease with elevated plasma ALP levels,

accompanied by bile duct obstruction, intrahepatic cholestasis or liver invasive disease (Krones et al., 2015). Bilirubin metabolism disorders and the abnormal structure or function of biliary system caused an increase in bilirubin (named jaundice), and serum bilirubin (TBIL and DBIL) is the most direct indicator for the detection of liver metabolic status (Fevry, 2008).

It has been proved that cholestasis was a disease caused by impaired balance among bile acid synthesis, conjugation, uptake and excretion (Ofliver, 2009). Cholestasis led to the intracellular retention of toxic bile constituents, subsequent liver injury, and ultimate biliary fibrosis and cirrhosis (Vries and Beuers, 2017). Therefore, the detection of TBA and bile flow rate is necessary for the diagnosis of liver diseases such as cholestasis. Intrahepatic cholestasis could cause the changes in vacuolar formation, hepatocyte necrosis and neutrophil infiltration in liver tissue through histopathological examination. Our results indicate that TJ treatment with fresh RS significantly reduced the elevated levels of serum enzymes (ALT, AST and ALP) and components (TBA, TBIL and DBIL), enhanced the flow of bile, alleviated the pathological damage and improved the structure of hepatocytes caused by ANIT.

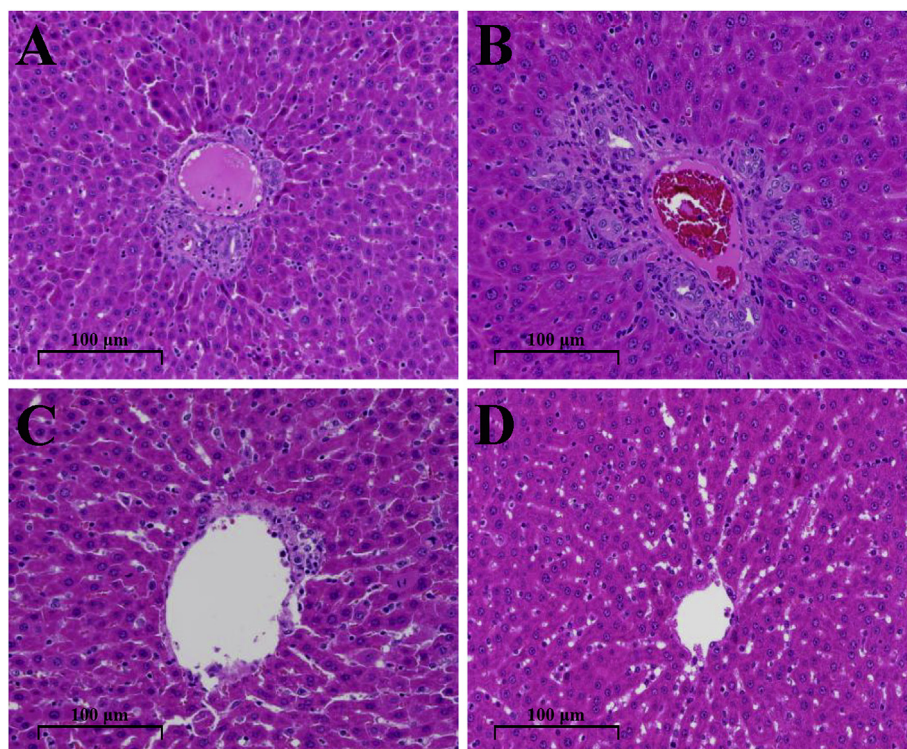


Fig. 7. Representative H&E stained sections of rat liver specimens in normal group (A), model group (B), TJ group (C) and UDCA group (D) at 108 h after 2% ANIT (80 mg/kg B.W.) administration. Scale bar = 100 μm, original magnification 200 $\times$.

In ANIT-induced intrahepatic cholestasis, oxidative stress accompanied with lipid peroxidation was involved in the development of cholestasis liver injury (Jin et al., 2013). MDA, the principal and most studied product of polyunsaturated fatty acid peroxidation, is a highly toxic molecule and should be considered as more than just a marker of lipid peroxidation. It was reported that the interaction of MDA with DNA and proteins was potentially mutagenic and atherogenic (Del Rio et al., 2005). In addition, MDA can be used as an indicator to reflect impaired membrane structure and functional integrity (Pizzimenti et al., 2013). NO, as a cell signaling molecule, maintained the homeostasis of hepatic microcirculation and exhibited a bidirectional effect on liver lesions (Chen et al., 2003). High reactivity of NO could scavenge peroxyl radicals and inhibited lipid peroxidation in vivo. However, NO reacted with peroxide to form peroxynitrite, which were harmful to the liver cells (Gutteridge and Halliwell, 2010). Our results show that the levels of MDA and NO increased significantly in model group and decreased obviously in TJ group, indicating that TJ treatment with fresh RS could protect liver cells from ANIT-induced liver injury.

5. Conclusions

In this paper, TJ treatment with fresh RS exhibited good therapeutic effect with significantly ameliorated liver injuries in ANIT-induced intrahepatic cholestasis of rats, which may be attributed to the recovery of oxidative damage in liver tissue. The results of phytochemical analysis by GC-MS show that lactones, flavonoids and phenolic acids were the main chemical constituents, with the content of irritant proto-anemonin in fresh RS about 25-fold higher than that in dried RS. It is rational that fresh RS was commonly chosen and applied to the TJ herbal patches rather than dried one. Therefore, TJ therapy with fresh RS has great potential in the treatment of cholestasis jaundice, especially for the patients who cannot be orally administered. However, the underlying mechanism of TJ treatment with fresh RS is still under further investigation.

Author contributions

Z.Z. designed the project. Z.Z., Y.M. and M.X. carried out the experimental work and writing of original draft. W.C. helped to direct the project and measured the chemical constituents of RS by gas chromatography-mass spectrometer. C.Y., X.S. and Q.G. helped to complete the experimental work. Q.Z. designed and directed the project and wrote the manuscript.

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Glossary

ALP: alkaline phosphatase
ALT: alanine aminotransferase
ANIT: α -naphthylisothiocyanate
AST: aspartate aminotransferase
DBIL: direct bilirubin
H&E: hematoxylin and eosin
GC-MS: chromatography-mass spectrometer
MDA: malondialdehyde
NO: nitric monoxide
RS: *Ranunculus sceleratus* L.
SD: standard deviation
TBA: total bile acid
TBIL: total bilirubin
TCM: traditional Chinese medicine
TJ: TianJiu
UDCA: ursodeoxycholic acid.

Update

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Corrigendum

Corrigendum to “TianJiu therapy for α -naphthyl isothiocyanate-induced intrahepatic cholestasis in rats treated with fresh *Ranunculus sceleratus* L.” [J. Ethnopharmacol. 248 (2020) 112310]

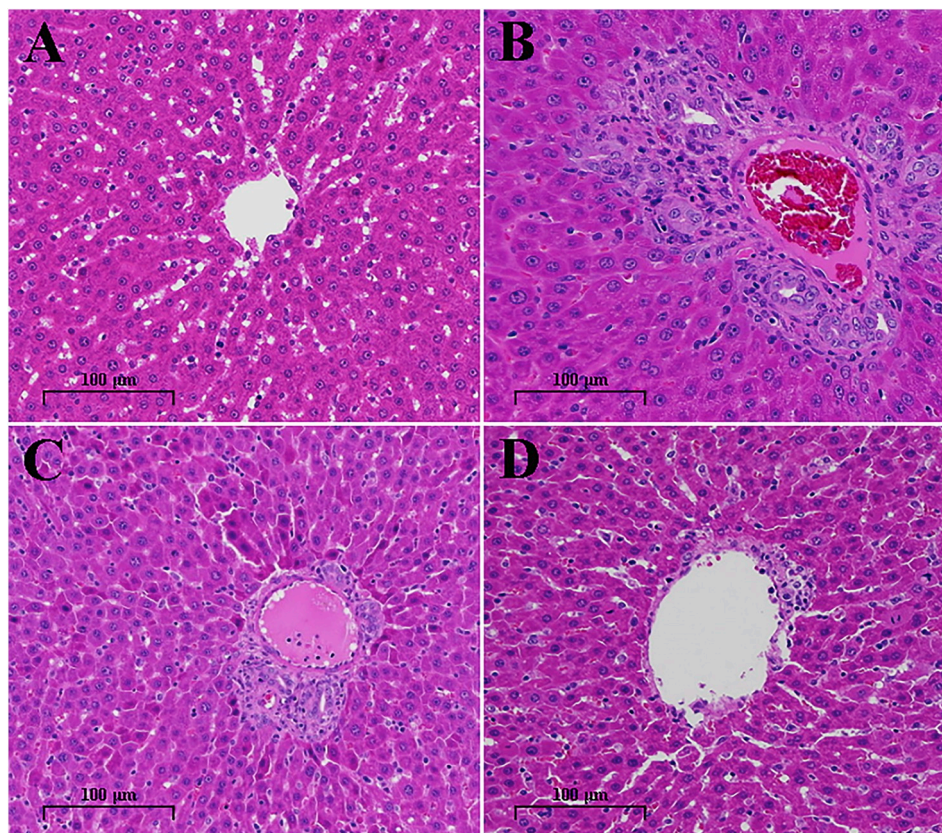
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The authors regret that Fig. 7 is incorrect. The correct figure is provided below:

The authors would like to apologise for any inconvenience caused.



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